

# Traceable Reproduction Audit and Feedback-Assisted Diagnostics for VLMPC-Style Language-Table Pushing

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**Abstract**—Aggregate success rates obscure which component actually controls a vision-language model predictive control (VLMPC) system. We introduce an execution-authority audit for Language-Table pushing that records the requested and selected targets, state source, raw planner action, executed action, and terminal outcome. The audit localizes the dominant tested gap: the raw video-prediction chain reaches 0/16 successes, while the same semantic controller and seeds reach 6/6 with oracle state and 0/6 with detector/tracker feedback; the paired visual-minus-oracle final-distance gap is  $0.2548 \pm 0.1453$ . In grounded diagnostics, feedback replaces 23/23 raw actions in every audited run, exposing the transfer of execution authority. Independent no-cache evaluations obtain 9/9 language matches and joint controls, 8/9 target-image matches and joint active successes, and 6/6 semantic repairs with 5/6 true-target completions under forced re-query versus 0/6 without re-query. These component-level comparisons separate grounding, planning, feedback estimation, and recovery, turning a reproduction outcome into a traceable diagnosis.

**Keywords**—Vision-language model predictive control, Language-Table, reproduction audit, feedback control, event re-query.

## I. INTRODUCTION

Vision-language manipulation couples a semantic question—what object does the instruction identify?—with a control question—how should that object be moved? Model predictive control (MPC) supplies receding-horizon action selection [1], [2], while vision-language models (VLMs) support task grounding [3]–[5]. VLMPC combines these capabilities by using semantic evaluation within a visual-prediction control loop [6]; trajectory-level variants reduce the cost of full video prediction [7].

The resulting pipeline is powerful but difficult to diagnose. A correct target can be paired with an ineffective controller, a successful trajectory can be produced after the raw planner action is replaced, and a re-query can fire without changing an already successful outcome. An aggregate task score cannot distinguish these cases. This ambiguity is especially consequential in reproduction studies: without an explicit record of state source and executed action, success may be assigned to a component that did not control the environment.

We address this attribution problem with an execution-authority audit. Every branch records the requested target, selected target, control-state source, raw planner action, executed action, and target-relative outcome. Matched comparisons then isolate four questions: whether the raw visual planner reaches the task criterion; whether a feedback law replaces its actions; whether oracle and visual state estimates support the same semantic controller; and whether language, target-image grounding, and re-query remain effective when evaluated independently.

The resulting evidence identifies a feedback-source bottleneck rather than a semantic one. The raw chain is 0/16, and grounded logs show that state feedback replaces 23/23 raw actions in every audited run. With the semantic controller and seeds fixed, oracle state yields 6/6 successes and detector/tracker feedback yields 0/6. Meanwhile, strict no-cache language trials reach 9/9 target matches, the target-image matrix reaches 8/9, and forced re-query repairs 6/6 injected semantic faults. The oracle-state branches are privileged diagnostics, not successful reproductions of the unmodified VLMPC chain. The contribution is therefore not another aggregate controller score, but a reusable protocol for assigning component responsibility and a file-traceable evaluation that exposes where the tested system succeeds and where the closed loop breaks.

## II. RELATED WORK

Visual MPC uses action-conditioned prediction to control manipulation directly from images [8], [9]. Language-conditioned systems separate semantic and spatial reasoning in different ways: CLIPort decomposes semantic and spatial reasoning [10], SayCan combines language plans with affordances [3], Inner Monologue incorporates feedback [5], and VoxPoser constructs spatial value maps [11]. These systems motivate an explicit connection between high-level selection and low-level execution.

Evaluation work has also moved beyond isolated demonstrations. VLATest probes robustness [12], while AHA structures failure reasoning [13]. Our audit complements these ap-

TABLE I  
RECOVERABLE IMPLEMENTATION PARAMETERS FROM CODE, METRICS,  
AND SUPPLEMENT CSVs.

| Parameter         | Value or rule used in the combined audit                                                                                                                         |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Environment       | BLOCK_8 Language-Table, 10 Hz; fresh raw sweep uses seeds 45–48; semantic, target-image, and event controls use 42–44; visual feedback uses 45–50                |
| Raw MPC           | action dim. 2, horizon 20; sampled candidates clipped to $[-0.02, 0.02]$ , executed action/feedback bound $[-0.08, 0.08]$                                        |
| Budgets           | default 200/5 samples/plan frequency; eight archived ablations: pf1/ns5, pf2/ns5, pf2/ns10, pf3/ns10, pf2/ns15, cost history low, cost default, cost target high |
| Detector/tracker  | YOLO conf. 0.5; PySOT SiamRPN tracker                                                                                                                            |
| Cost weights      | contact/approach/alignment/obstacle = 0.35/0.15/18.0/24.0                                                                                                        |
| Progress terms    | reward 1.2, non-progress penalty 12.0, min gain 1.0, obstacle decay 42 px                                                                                        |
| Close range       | distance 135 px; contact/alignment/reward/penalty = 0.10/6.0/1.8/20.0                                                                                            |
| Feedback push     | contact offset 0.06, clearance 0.12, move step 0.06, push step 0.075, tolerance 0.018                                                                            |
| Original event    | default window/min-progress/min-step = 3/1.0/3; semantic matched controls use 4/0.002/3 with min progress in world-distance units                                |
| Semantic event    | window 4, min progress 0.002 world-distance units, distance floor $1.5 \times$ success distance                                                                  |
| Online VLM        | CodexCLIClient; local codex exec -m gpt-5.5; isolated CODEX_HOME; credentials excluded                                                                           |
| Success threshold | evaluation tolerance $d_{\text{world}} \leq 0.08$ ; strict audit also reports 0.05 and 0.10                                                                      |

proaches by treating state source and executed-action authority as first-class evaluation variables. It is designed to answer not only whether a run succeeds, but which module supplies the information and action that produce its outcome.

### III. EXECUTION-AUTHORITY AUDIT

#### A. Task, Metric, and Evidence

The experiments use Language-Table desktop pushing. A fixed label, language instruction, target image, or scene-selection request identifies an object to be pushed toward a designated corner. Success is measured by the final world-coordinate target-to-goal distance,

$$\text{success} = \mathbb{I}[d_{\text{world}}^{\text{final}} \leq 0.08]. \quad (1)$$

The archived 0.08 tolerance is the primary criterion; 0.05 is also reported to distinguish completion from tighter placement. All counts retain their own experimental denominator. In particular, the 18-run semantic suite measures control of the selected target, whereas the independent nine-run language and target-image studies measure grounding against labels specified outside the controller.

Each run is represented as  $(g_{\text{req}}, g_{\text{sel}}, s_t, a_t^{\text{raw}}, a_t^{\text{exec}}, y)$ : requested and selected targets, control-state source, raw action, executed action, and terminal outcome. The repository contains per-run CSVs, exact parameter records, source and asset manifests, figure scripts, and an automated consistency check. Thus the compact tables below summarize, rather than replace, the run-level evidence.

#### B. Attribution Branches

Figure 1 shows the common audit path. The *raw* branch grounds a target, samples candidate sequences, predicts visual futures, ranks them, and executes  $a_t^{\text{raw}}$ . Its horizon is 20 with two-dimensional actions; the default budget is 200 candidate

samples with replanning every five steps. The fresh sweep uses four controller variants and four seeds, and an archived ablation covers eight sampling/cost configurations; exact costs, detector settings, event thresholds, and source hashes are supplied in the supplement.

The *grounded* branch preserves and logs the raw planner call but executes a bounded simulator-state correction. The feedback law stages the end effector, establishes contact, and pushes toward the goal; the supplement provides its gains and phase thresholds. Logging both actions makes authority observable: all 12 grounded runs contain 23 raw actions and 23 replacements.

The *semantic* branch accepts fixed labels, instructions, target images, or VLM scene selection and applies the same state-feedback controller to the selected target. It retains receding-horizon execution and logging, but no raw video-prediction sampling has action authority. The *visual-feedback* branch holds that semantic controller and the seeds fixed while replacing simulator state with YOLO/PySOT detector-tracker estimates. This paired change isolates feedback source.

Formal language, target-image, and re-query calls use the project’s CodexCLIClient with the GPT-5.5 model alias. Prompts are passed through standard input, images through CLI attachments, and non-ASCII paths are staged in an ASCII-only temporary directory. An isolated CODEX\_HOME keeps credentials outside the source tree. The reported language and event studies disable caching and record zero cache hits.

#### C. Re-query Intervention

Natural-event controls use a four-step stagnation window and a minimum progress of 0.002 world-distance units. Three event-enabled runs trigger a fresh query, while matched no-event runs retain identical final distances because both conditions already succeed. A separate intervention supplies the missing recovery condition: six seeds begin with the true red-moon target replaced by blue cube; treatment issues a forced audit query at step 3, whereas control retains the corrupted target. Evaluation always uses the independently specified true target.

## IV. EXPERIMENTS AND RESULTS

#### A. Evidence Design and Raw Reference

The study separates raw planning, execution authority, feedback source, grounding, and recovery. Raw planning uses four controller variants at seeds 45–48 plus eight archived configurations; the oracle-visual comparison fixes controller and seeds 45–50; language and target-image studies repeat independent requests at seeds 42–44. Tables III–VI and Figs. 2–3 keep these denominators and distributions visible.

#### B. Authority and Feedback Source

Raw variants remain above tolerance. Grounded logs expose the mechanism that changes the outcome: all 12 runs replace every raw action (23/23), while the paired state-source experiment yields 6/6 with oracle feedback and 0/6 with visual estimates, a per-seed gap of  $0.2548 \pm 0.1453$ .

TABLE II  
CONTROLLER BRANCHES USED TO ATTRIBUTE SEMANTIC ROLE, STATE SOURCE, AND EXECUTED-ACTION AUTHORITY.

| Branch                   | Semantic/planning role                                      | Actual MPC role                                                              | Executed-action authority                                            | State source        |
|--------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------|
| Unmodified VLMPC         | Ground target; sample sequences; predict visual futures     | Candidate ranking with a video-prediction cost; first raw action is executed | Raw planner action                                                   | Visual observations |
| Grounded original MPC    | Preserve and log the raw planner call                       | Raw planner retained for comparison and traceability                         | State-feedback correction replaces raw action (23/23 in all 12 runs) | Simulator state     |
| Semantic MPC             | Select from label, language, image, or scene query          | Receding-horizon execution/logging; no active raw video-prediction sampling  | State-feedback push law                                              | Simulator state     |
| Visual feedback          | Select a fixed target; estimate state from RGB              | Detector/tracker feedback under the same semantic controller                 | Visual feedback law                                                  | RGB observations    |
| Event-control supplement | Monitor stagnation or semantic events and refresh the query | Event logic controls query timing, not the low-level push law                | Semantic-MPC feedback action                                         | Simulator state     |

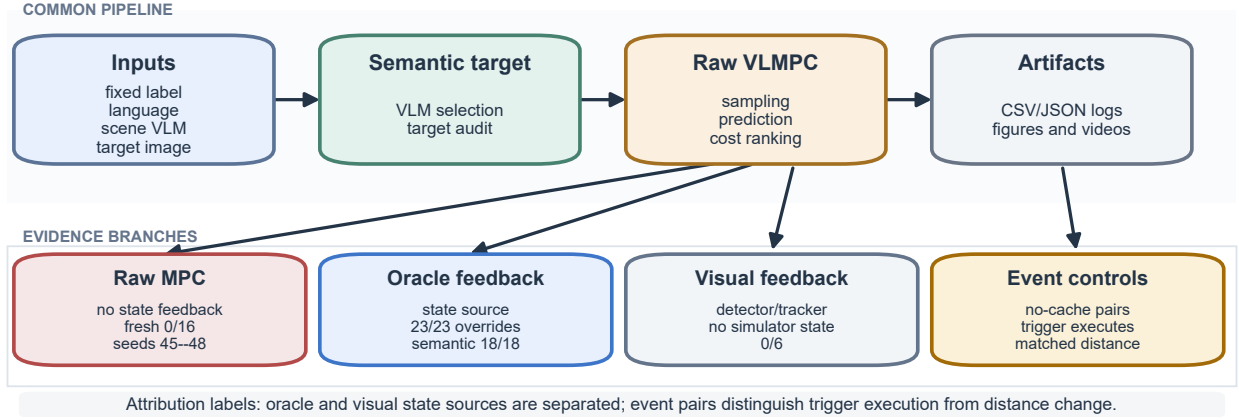


Fig. 1. Execution-authority audit. Explicit state-source and action-authority labels separate semantic selection, raw visual planning, feedback execution, and re-query recovery.

TABLE III  
RAW-CHAIN REFERENCE ACROSS SEEDS 45–48 AND EIGHT RECORDED PARAMETER CONFIGURATIONS.

| Variant/suite     | Runs | Succ. | Final dist.         | Range       |
|-------------------|------|-------|---------------------|-------------|
| Baseline          | 4    | 0/4   | $0.3358 \pm 0.1099$ | 0.199–0.436 |
| Improved          | 4    | 0/4   | $0.3354 \pm 0.1048$ | 0.224–0.436 |
| Literature        | 4    | 0/4   | $0.3134 \pm 0.1325$ | 0.163–0.436 |
| Literature-v2     | 4    | 0/4   | $0.3134 \pm 0.1325$ | 0.163–0.436 |
| All fresh raw     | 16   | 0/16  | $0.3245 \pm 0.1085$ | 0.163–0.436 |
| Archived ablation | 8    | 0/8   | $0.2534 \pm 0.0304$ | 0.216–0.294 |

At 0.05/0.08/0.10: fresh raw 0/16, 0/16, 0/16; archived ablation 0/8, 0/8, 0/8.

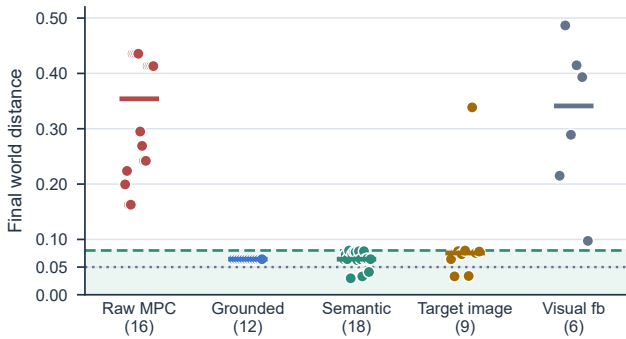


Fig. 2. Final-distance distributions across raw, grounded, semantic, target-image, and visual-feedback suites.

### C. Grounding and Recovery

Independent labels give 9/9 language matches. The target-image selector reaches 4/4 versus 1/4 for the earlier heuristic;

Threshold sensitivity

|                             |         |         |         |
|-----------------------------|---------|---------|---------|
| Fresh raw MPC (n=16)        | 0/16    | 0/16    | 0/16    |
| Archived ablation (n=8)     | 0/8     | 0/8     | 0/8     |
| Grounded diagnostics (n=12) | 0/12    | 12/12   | 12/12   |
| Semantic MPC (n=18)         | 3/18    | 18/18   | 18/18   |
| Target image (n=9)          | 2/9     | 8/9     | 8/9     |
| Visual feedback (n=6)       | 0/6     | 0/6     | 1/6     |
|                             | <= 0.05 | <= 0.08 | <= 0.10 |

Fig. 3. Threshold sensitivity at 0.05, 0.08, and 0.10. The outlined 0.08 column is the project criterion.

its active end-to-end matrix reaches 8/9. Event controls separate a trigger that executes from recovery that changes an erroneous semantic state. Tables VII–X and Fig. 4 retain the per-condition evidence.

## V. DISCUSSION

The audit turns a mixed full-pipeline outcome into a component-level diagnosis. Semantic grounding is strong across repeated language and target-image requests, and a fresh query can repair an injected semantic fault. Stable execution, however, depends on the state and action path: raw video prediction does not complete the task in the evaluated sweep, feedback replaces every raw action in the grounded

TABLE IV  
BRANCH-LEVEL RESULTS. STATE SOURCE REMAINS EXPLICIT SO ORACLE AND VISUAL ROWS ARE NOT POOLED.

| Suite                | Runs      | Control    | Evidence and semantic check             | Final dist.             | State source |
|----------------------|-----------|------------|-----------------------------------------|-------------------------|--------------|
| Grounded controller  | 4         | 4/4        | fixed target; 23/23 override            | 0.0643±0.0000           | oracle       |
| Grounded ablation    | 8         | 8/8        | fixed target; 23/23 override            | 0.0643±0.0000           | oracle       |
| Event controls       | 9         | 9/9        | matched; natural trigger 3/3 enabled    | 0.0587±0.0233 matched   | oracle       |
| Semantic MPC live    | 18        | 18/18*     | 15 fixed; one each language/image/scene | 0.0641±0.0150           | oracle       |
| Target-image matrix  | 9         | 8/9 joint  | semantic 8/9; no pre-satisfied          | 0.0950±0.0932           | oracle       |
| Oracle feedback      | 6         | 6/6        | fixed target; seeds 45–50               | 0.0612±0.0206           | oracle       |
| Visual feedback      | 6         | 0/6        | same controller and seeds               | 0.3160±0.1439           | visual       |
| Semantic-fault audit | 12 paired | 5/6 treat. | repair 6/6; control 0/6                 | 0.0639±0.0157 treatment | oracle       |
| Language repeats     | 9         | 9/9 joint  | 3 instructions × 3 seeds; no cache      | 0.0661±0.0191           | oracle       |

\*Selected-target control. Independent language and target-image evaluations provide grounding denominators.

TABLE V  
CONDITION-WISE VARIABILITY FOR THE SEMANTIC-MPC SUITE (MEAN ± SAMPLE SD).

| Condition         | <i>n</i> | Mean ± SD       | ≤ 0.05 | ≤ 0.08 | Target/seed scope      | Interpretation          |
|-------------------|----------|-----------------|--------|--------|------------------------|-------------------------|
| Input forms       | 4        | 0.0663 ± 0.0041 | 0/4    | 4/4    | 3 forms + target image | Input-form comparison   |
| Multi-target      | 8        | 0.0678 ± 0.0167 | 1/8    | 8/8    | 8 target objects       | Small target sweep      |
| Multi-seed        | 3        | 0.0587 ± 0.0233 | 1/3    | 3/3    | red moon, seeds 42–44  | Seed variability        |
| Semantic ablation | 3        | 0.0568 ± 0.0136 | 1/3    | 3/3    | default/precise/fast   | Controller settings     |
| All latest runs   | 18       | 0.0641 ± 0.0150 | 3/18   | 18/18  | heterogeneous union    | Selected-target control |

TABLE VI  
PER-RUN DIGEST EXTRACTED FROM ARCHIVED AND LATEST CSV FILES. EXEC. S USES  $d_{\text{world}}^{\text{final}} \leq 0.08$ .

| Suite                | Final-distance summary                     | Range        | Exec. S       | Semantic/VLM evidence          |
|----------------------|--------------------------------------------|--------------|---------------|--------------------------------|
| Fresh raw MPC        | 4 variants × 4 seeds; 0.3245 ± 0.1085      | 0.163–0.436  | 0/16          | raw visual chain               |
| Archived raw         | eight parameter settings                   | 0.216–0.294  | 0/8           | archived comparison            |
| Grounded controller  | all 0.064                                  | 0.064–0.064  | 4/4           | fixed target                   |
| Grounded ablation    | all 0.064                                  | 0.064–0.064  | 8/8           | fixed target                   |
| Event controls       | matched event/no-event distances identical | 0.033–0.079  | 9/9           | natural trigger 3/3            |
| Semantic MPC live    | min 0.030; median 0.064; max 0.0798        | 0.030–0.0798 | 18/18         | 15 fixed; language/image/scene |
| Target-image matrix  | one yellow-pentagon error                  | 0.033–0.339  | 8/9 joint     | all active trials              |
| Visual feedback      | detector/tracker, seeds 45–50              | 0.097–0.487  | 0/6           | no simulator state             |
| Fault revision audit | treatment repairs 6/6; control 0/6         | 0.044–0.385  | 5/6 treatment | forced re-query                |
| Language repeats     | 3 instructions × seeds 42–44               | 0.034–0.0799 | 9/9           | no cache                       |

TABLE VII  
STRICT NO-CACHE EVENT RE-QUERY CONTROLS.

| Case                            | Natural query   | Final dist.   | Note                    |
|---------------------------------|-----------------|---------------|-------------------------|
| No-event, seeds 42–44           | 0               | 0.0332–0.0787 | 3/3; 3 fresh responses  |
| Event-enabled, seeds 42–44      | 1 each          | 0.0332–0.0787 | 3/3; 6 fresh responses  |
| Forced scene audit, seeds 42–44 | 1/3; forced all | 0.0339–0.0725 | 3/3; 14 fresh responses |

TABLE VIII  
CONTROLLED SEMANTIC-FAULT RE-QUERY AUDIT.

| Condition            | <i>n</i> | Repair | True success | Final dist.     |
|----------------------|----------|--------|--------------|-----------------|
| No-query control     | 6        | 0/6    | 0/6          | 0.2435 ± 0.0904 |
| Forced query (Codex) | 6        | 6/6    | 5/6          | 0.0639 ± 0.0157 |

TABLE IX  
TARGET-IMAGE SEMANTIC-SELECTION AUDIT.

| Case              | Expected     | Old       | VLM          | OK  |
|-------------------|--------------|-----------|--------------|-----|
| Red-moon crop     | red moon     | blue cube | red moon     | yes |
| Blue-cube crop    | blue cube    | blue cube | blue cube    | yes |
| Yellow-pent. crop | yellow pent. | blue cube | yellow pent. | yes |
| Archived frame    | red moon     | blue cube | red moon     | yes |

diagnostics, and the same semantic controller separates sharply under oracle and visual feedback. This is precisely the distinction that aggregate success rates erase.

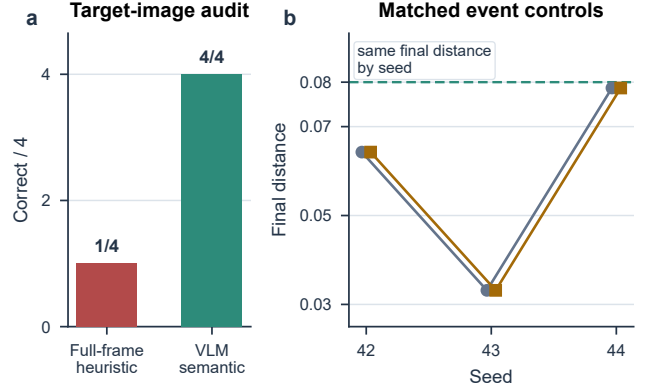


Fig. 4. Target-image and event audits: selector accuracy and matched no-cache event controls.

The inference is scoped to Language-Table, the listed assets, and the evaluated controller variants. Oracle-feedback runs diagnose controllability; they are not evidence that the raw visual planner succeeded. Likewise, the 18/18 semantic-suite count measures control of the selected target, while the nine-run language and target-image matrices supply independent grounding denominators. The repeated language and target-image studies use three seeds per condition, and the heterogeneous 18-row suite is not interpreted as 18 independent seeds; the grounded zero variance reflects its shared terminal feed-

TABLE X  
REPEATED TARGET-IMAGE END-TO-END MATRIX. ALL TRIALS BEGIN OUTSIDE THE 0.08 THRESHOLD.

| Case            | Runs | VLM match | Pre-satisfied | Active joint | Final distance      | Interpretation         |
|-----------------|------|-----------|---------------|--------------|---------------------|------------------------|
| Red moon        | 3    | 3/3       | 0             | 3/3          | $0.0587 \pm 0.0233$ | All trials succeed     |
| Blue cube       | 3    | 3/3       | 0             | 3/3          | $0.0623 \pm 0.0249$ | All trials succeed     |
| Yellow pentagon | 3    | 2/3       | 0             | 2/3          | $0.1640 \pm 0.1512$ | Seed-42 semantic error |
| All cases       | 9    | 8/9       | 0             | 8/9          | $0.0950 \pm 0.0932$ | One semantic miss      |

TABLE XI  
FACTOR-ISOLATING BOTTLENECK EVIDENCE.

| Factor              | Evidence and implication                                                  |
|---------------------|---------------------------------------------------------------------------|
| Raw planning        | Fresh 0/16 and archived 0/8: tested variants do not reach completion.     |
| Execution authority | Grounded success follows 23/23 action replacement in every run.           |
| Feedback source     | Fixed controller/seeds: oracle 6/6, visual 0/6; gap $0.2548 \pm 0.1453$ . |
| Semantic grounding  | Language 9/9; target-image 8/9; retained miss is localized.               |

TABLE XII  
ROBUSTNESS-EDGE AUDIT FOR DETECTION AND UNKNOWN-TARGET HANDLING.

| Case             | Status      | Confidence |
|------------------|-------------|------------|
| Original         | ok          | 0.902      |
| Occluded target  | ok          | 0.632      |
| Viewpoint proxy  | ok          | 0.902      |
| Brightness shift | ok          | 0.900      |
| Purple pole      | safe reject | —          |
| Green moon       | safe reject | —          |
| Unknown target   | safe reject | —          |

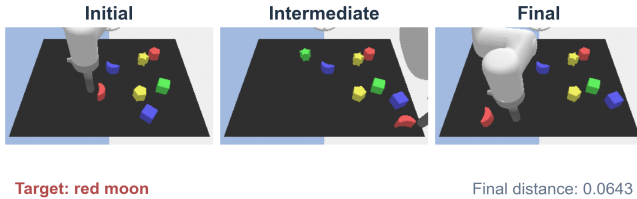


Fig. 5. Representative frames from a logged oracle-feedback red-moon execution.

back law. The factor-isolating audit localizes the tested boundary without claiming exhaustive causal decomposition. Input-level occlusion, brightness, viewpoint-proxy, and unknown-target tests are retained in the supplement as robustness checks rather than presented as real-world generalization.

The measured oracle-to-vision gap also defines the next experiment. Synchronized RGB/state traces should quantify target-centroid error, end-effector error, goal-distance mean absolute error, visibility F1, and temporal consistency before repeating the matched oracle-versus-vision evaluation. Confidence-gated updates can then be added without changing the semantic interface, preserving the same attribution design. Recovery can be extended with randomized fault times and matched no-query, forced-query, and natural-event conditions.

#### A. Data and Code Availability

The public repository contains the manuscript source, authoritative per-run and aggregate CSVs, exact parameters, figure-generation code, source/runtime manifests, and an auto-

mated consistency check. Credentials, private endpoints, local paths, raw service logs, and third-party weights are excluded; included aggregates can be recomputed from the portable files. The robustness-edge cases in Table XII and representative execution in Fig. 5 are supplied as portable supplementary evidence, not as claims of real-world generalization.

## VI. CONCLUSION

We introduced an execution-authority audit that separates target selection, state source, raw planning, executed action, and task outcome. Matched evidence identifies feedback estimation and action authority as the dominant tested bottleneck, while independent evaluations verify repeated grounding and semantic-fault recovery. The protocol provides a traceable basis for assigning component responsibility before aggregate VLMPC results are interpreted.

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